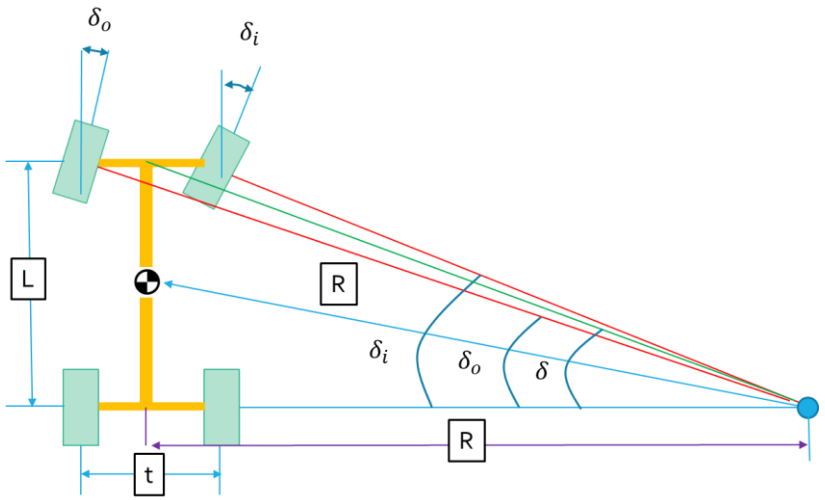
Example Problem on Ackermann angle

Find the ideal Ackermann steer angle for inner and Outer wheel when a vehicle with wheelbase of 2700 mm and Wheel track of 1450 mm is turning with a Turn circle radius of 4 m

Outer wheel $tan\delta_o = \dfrac{L}{R + t/2}$

Inner wheel $tan\delta_i = \dfrac{L}{R - t/2}$

$$tan\delta_o = \frac{2.7}{4 + 1.45/2} \qquad \delta_o = 29.7°$$

$$tan\delta_i = \frac{2.7}{4 - 1.45/2} \qquad \delta_i = 39.5°$$